

Question	Marking Guidance	Mark	Comments
02.1	1. Change in (sequence of) amino acid(s)/primary structure; 2. Change in hydrogen/ionic/disulfide bonds; 3. Alters <u>tertiary</u> /3 ⁰ structure;	3	1. Reject amino acids are formed. 1. Reject amino acids code. 3. Reject active site. 3. Ignore quaternary. 3. Ignore 3D.
02.2	1. Produce healthy (red blood) cells OR Produce (normal) polypeptide/haemoglobin; 2. No sickle/faulty/SCD (red blood) cells (produced) OR No defective polypeptide/haemoglobin; 3. Stem/marrow cells (continuously) divide/replicate OR Less chance of rejection (from brother/sister);	3	1. Produce only healthy (red blood) cells is only equivalent to mark point 1. 1. Accept produce 'normal'/non-SCD cells. 1. and 3. Ignore type of stem cell e.g. pluripotent. 3. Differentiate is not equivalent to divide/replicate.
02.3	(For gene therapy) 1. No destruction of bone marrow OR No destruction of stem cells; 2. Donors are not required; 3. Less/no chance of rejection (own stem cells); (Against gene therapy) 4. Sickle/faulty (red blood) cells still produced 5. Immune response against genetically modified cells/virus OR Long-term effect not known (as is new treatment) OR Virus could cause side effects;	3 max	Max 2 marks for marking points 1, 2 and 3 1. Accept no destruction of faulty bone marrow unless context indicates this is against gene therapy. 2. Stating 'only own cells used' is not equivalent. 5. Accept 'virus could cause problems' or 'risk(s) with virus'.
